THE STACKCALC LIBRARY • CURRICULUM SERIES — BOOK 4 OF 4

STACK SPEED & FLUENCY

15 Progressive Timed Sprints for Inside-Out Mathematical Evaluation on the Four-Line Stack

A progressive Grades 3–12 timed fluency and competition workbook for the four-line StackCalc SC-6 Elementary and SC-32 Scientific calculators. Modeled after UIL Calculator Applications and Number Sense contests, these 15 sprints train students to decompose multi-step expressions from the inside out across Stack Levels 1, 2, and 4—with zero parentheses.

StackCalc

stackcalc.io

First Edition

BOOKS, BLOCKS & INSTRUMENTS IN THE STACKCALC LIBRARY

Illustrated Companion Trade Books (History of Mathematics & Stack Thinking)

- Volume I — From Blocks to the Stack: How Hands-On Blocks Make Numbers Click (Grades K–6 / StackCalc SC-6)
- Volume II — Thinking in the Stack: The Art of Calculating Without Parentheses (Grades 7–12, AP & Engineers / StackCalc SC-32)

K–12 Concept-First Curriculum & Student Workbooks (US Letter 8.5" × 11" Editions)

- Book 1 — Teaching with the Stack: Unified K–12 Master Teacher's Guide, Concept Lessons & Complete Answer Keys
- Book 2 — Hands-On Math Lab: K–12 Student Manipulative & Four-Line Stack Exploration Workbook
- Book 3 — Middle School Math Lab: Grades 6–8 Two-Method Student Practice Workbook (900 Problems) & 15-Page Coach Guide
- Book 4 — Stack Speed & Fluency: 15 Progressive Timed Sprints for Inside-Out Mathematical Fluency

Three Published Physical Manipulative Block Studios & Four-Line Stack Calculators (stackcalc.io)

- Place Value Studio (PLACE-01): 6×2 Whole & Decimal Trays, Digit Blocks (0–9) & Stackable Gold [+1] Carry Tokens
- Fraction Studio (FRAC-01): 84.0 mm Proportional Fraction Towers (1 Whole to 1/12) & 12-Socket Comparator Lid
- Prime Factor Studio (PRIME-01): Logarithmic Prime Blocks (2, 3, 5, 7 where equal products stand flush) & GCF/LCM Comparator
- StackCalc Calculators: SC-6 Elementary/Middle School Calculator & SC-32 Scientific Calculator (Hardware, iOS, iPadOS, watchOS & Web)

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STACK SPEED & FLUENCY: 15 Progressive Timed Sprints for Inside-Out Mathematical Evaluation on the Four-Line Stack

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State & District Curriculum Alignment:

Designed to supplement state-mandated mathematics standards—including Common Core State Standards (CCSS-M), Texas Essential Knowledge and Skills (TEKS), Florida B.E.S.T., Virginia SOL, and AP Mathematics / Computer Science—alongside district basal textbooks (Eureka Math / EngageNY, Illustrative Mathematics / Open Up, enVision Mathematics, Big Ideas Math, HMH Into Math, McGraw-Hill Reveal, Bridges in Mathematics, and Singapore Dimensions Math).

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Physical calculator instruments that utilize coin or button cell batteries must maintain a captive, tool-secured battery compartment compliant with UL 4200A. 3D-printed manipulative blocks constitute small parts under 16 CFR Part 1501; adult supervision is required for children under 6 years of age.

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Preface: How Your Four-Line Stack Calculator Works

Read this 3-minute introduction before starting Trial 1—no prior RPN calculator experience required!

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1. WHY ENGINEERS & MATH COMPETITORS USE A STACK CALCULATOR WITH NO [=] KEY

In timed mathematics competitions—such as UIL Calculator Applications—and in real-world engineering, the #1 cause of wrong answers on a standard calculator is lost or mismatched parentheses. When you type $((14 + 6) \times 3) \div (9 - 4) =$ left-to-right on a 1-line screen, a single missing parenthesis silently ruins the entire calculation.
Your StackCalc calculator (SC-6 for Grades K–8; SC-32 Scientific for Grades 7–12+) solves this by replacing parentheses with four visible memory shelves called the Stack: Floor 1 (Line X, your active Workbench), Floor 2 (Line Y, your Waiting Shelf), Floor 3 (Line Z), and Floor 4 (Line T). Just like solving on paper, you evaluate the innermost group first, let it wait visibly on Shelf Y while you build the next group on Workbench X, and press the operation key (+, −, ×, ÷) to combine them!

2. WARM UP IN 90 SECONDS: TRY THESE THREE STACK PATTERNS RIGHT NOW

Pattern A — Two-Number Operation (Level 2 Stack): Evaluate $24 + 18 = 42$

Type [24] and press [ENTER] to push 24 onto Waiting Shelf Y. Type [18] on Workbench X, then press [+] → 42!

Press Keys: 

Pattern B — Chained Two-Group Expression Without Parentheses: $(14 + 6) \div (9 - 4) = 4$

Solve the numerator first: [14] [ENTER] [6] [+] = 20. Then just start typing [9] [ENTER] [4] [-]! 20 automatically lifts to Y while 5 forms on X. Press [÷] → 4!

Press Keys: 

Pattern C — Instant Doubling & Squaring Without Retyping: $2 \times 37 = 74$

Because pressing [ENTER] copies Line X into Line Y, [37] [ENTER] [+] doubles 37 in three keystrokes! (And [37] [ENTER] [×] squares it!)

Press Keys: 

3. HOW THIS WORKBOOK CONNECTS TO OUR CURRICULUM & BLOCK KITS

• 100% Self-Contained Fluency Sprints: You do not need to have read any other book to use this workbook. Each sprint focuses on inside-out mathematical evaluation—writing your paper setup or first inner step, computing the answer, and checking on your calculator.
• Aligned to Books 1–3: Fluency Checks 1–4 & C1 build K–2 number sense (+5 points per item, no penalty for mistakes). Time Trials 5–12 & C2–C3 build Grades 3–12 fluency across Fractions, Multi-Step Expressions, Quadratics, Trigonometry, Exponentials, Combinatorics & Binary/Hex Logic.

Stack Speed & Fluency Trials

Progressive mathematical fluency checks (Units 1–4) and UIL-style timed sprints (Units 5–12).

1. FLUENCY CHECK (GRADES K–2) & TIME TRIAL (GRADES 3–12) SCORING RULES

Trials 1–4 and C1 are Foundational Fluency Checks (+5 points per correct answer, no penalty for mistakes). Trials 5–12 and C2–C3 are modeled after UIL Calculator Applications and Number Sense contests (+5 points correct, −2 points error) to build inside-out mathematical accuracy and verification speed.

CORRECT ANSWER	INCORRECT / ERROR	UNATTEMPTED / SKIP	SPEED BONUS
+5 Points	**-2 Points**	**0 Points**	**+1 pt / rem. min**

2. MAXIMUM STACK LEVEL DISCIPLINE

Competitors must respect the Max Stack Depth badge on each trial: Level 1 uses only Register X; Level 2 uses Registers X and Y; Level 4 uses the full 4-register stack (T, Z, Y, X). Pressing unnecessary [ENTER] keys that overflow the allowed depth is penalized.

3. OFFICIAL COMPETITOR TIME TRIAL LOG SHEET

Trial	Mathematical Topic	Max Stack	Time	Attempt	Correct	Errors	
T-1	Unit 1 Fluency Check: Step Counting & One-More Addition	**Level 1**	5 min	___/10	___(+5)	___(-2)	_____
T-2	Unit 2 Fluency Check: Part-Part-Whole Number Bonds	**Level 2**	5 min	___/10	___(+5)	___(-2)	_____
T-3	Unit 3 Fluency Check: Complements to 10 & Make-10 Bridging	**Level 2**	5 min	___/10	___(+5)	___(-2)	_____
T-4	Unit 4 Fluency Check: Doubles, Near-Doubles & Equal Groups	**Level 2**	5 min	___/10	___(+5)	___(-2)	_____
T-CUM_1	Cumulative Check C1: Primary Math Foundations (Units 1–4)	**Level 2**	10 min	___/10	___(+5)	___(-2)	_____
T-5	Unit 5 Time Trial: Multi-Step Operations & Inner Groups	**Level 4**	10 min	___/10	___(+5)	___(-2)	_____
T-6	Unit 6 Time Trial: Fractions, Mixed Numbers & Reciprocals	**Level 4**	10 min	___/10	___(+5)	___(-2)	_____
T-7	Unit 7 Time Trial: Nested Expressions & Chained Branches	**Level 4**	10 min	___/10	___(+5)	___(-2)	_____
T-8	Unit 8 Time Trial: Polynomials & Quadratic Factoring	**Level 4**	10 min	___/10	___(+5)	___(-2)	_____
T-CUM_2	Cumulative Trial C2: Intermediate Mathematics Championship	**Level 4**	15 min	___/10	___(+5)	___(-2)	_____
T-9	Unit 9 Time Trial: Trigonometry, Vectors & Polar Coordinates	**Level 4**	10 min	___/10	___(+5)	___(-2)	_____
T-10	Unit 10 Time Trial: Exponential Growth & Logarithms	**Level 4**	10 min	___/10	___(+5)	___(-2)	_____
T-11	Unit 11 Time Trial: Combinatorics, Statistics & Probability	**Level 4**	10 min	___/10	___(+5)	___(-2)	_____
T-12	Unit 12 Time Trial: Digital Logic, Base Conversions & CS	**Level 4**	10 min	___/10	___(+5)	___(-2)	_____
T-CUM_3	Cumulative Trial C3: Grand Mathematics & Stack Invitational	**Level 4**	20 min	___/10	___(+5)	___(-2)	_____

How to Run Weekly 5-Minute Friday Fluency Sprints

Students write their paper setup or inner step first, compute the answer, and verify on their SC-6 or SC-32 calculator (never writing out button lists). Check answers against the Self-Check Key (pp. 17–18).

Unit 1 Fluency Check: Step Counting & One-More Addition

Foundations of Counting, Cardinality & Unit-Step Addition • Focus: Place Value Studio (Digit Blocks 0–9 & Gold [+1] Step Tokens)

MAX STACK DEPTH: 1 | **TIME LIMIT: 5 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: _____:_____ **Score: _____ / 50**

1	**Count 3 unit steps: 1 + 1 + 1** Strategy Hint: Add 1 for each unit step	**Paper Setup / Inner Step:**	**Final Answer:**
2	**Start at 5, count on 2 more: 5 + 1 + 1** Strategy Hint: Count on from 5 by unit steps	**Paper Setup / Inner Step:**	**Final Answer:**
3	**Count 4 unit steps: 1 + 1 + 1 + 1** Strategy Hint: Four unit steps from 0	**Paper Setup / Inner Step:**	**Final Answer:**
4	**Count 8 unit steps from 0** Strategy Hint: Eight unit steps make 8	**Paper Setup / Inner Step:**	**Final Answer:**
5	**Start at 6, count back 2 steps: 6 - 1 - 1** Strategy Hint: Count back 1 at a time	**Paper Setup / Inner Step:**	**Final Answer:**
6	**Count 5 unit steps: 1 + 1 + 1 + 1 + 1** Strategy Hint: Five unit steps make 5	**Paper Setup / Inner Step:**	**Final Answer:**
7	**Skip-count by 2s three times: 2 + 2 + 2** Strategy Hint: Three equal steps of 2	**Paper Setup / Inner Step:**	**Final Answer:**
8	**Start at 7, count on 3 steps: 7 + 1 + 1 + 1** Strategy Hint: Count up to make a ten	**Paper Setup / Inner Step:**	**Final Answer:**
9	**Adding to zero: 0 + 9** Strategy Hint: Identity property of zero	**Paper Setup / Inner Step:**	**Final Answer:**
10	**Count back 3 steps from 10: 10 - 1 - 1 - 1** Strategy Hint: Count back from 10	**Paper Setup / Inner Step:**	**Final Answer:**

Foundational Fluency Check (No-Penalty Primary Scoring)

Award +5 points for each correct answer (0 points for skipped or incorrect items—no negative penalty). Build or solve mentally/on paper first, then check with your partner or teacher on the SC-6!

Unit 2 Fluency Check: Part-Part-Whole Number Bonds

Composing & Decomposing Quantities with Two-Row Number Bonds • Focus: Place Value Studio (Two-Row Tray & Digit Blocks)

MAX STACK DEPTH: 2 | **TIME LIMIT: 5 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: _____:_____ **Score: _____ / 50**

1	**Part 4 and Part 3 → Whole?** Strategy Hint: Combine the two parts (4 + 3)	**Paper Setup / Inner Step:**	**Final Answer:**
2	**Part 5 and Part 5 → Whole?** Strategy Hint: Equal parts that make 10	**Paper Setup / Inner Step:**	**Final Answer:**
3	**Whole 9, Part 6 → Missing Part?** Strategy Hint: Subtract known part from whole (9 - 6)	**Paper Setup / Inner Step:**	**Final Answer:**
4	**Whole 10, Part 4 → Missing Part?** Strategy Hint: Subtract known part from 10	**Paper Setup / Inner Step:**	**Final Answer:**
5	**Part 8 and Part 7 → Whole?** Strategy Hint: Combine parts across ten	**Paper Setup / Inner Step:**	**Final Answer:**
6	**Whole 14, Part 8 → Missing Part?** Strategy Hint: Decompose 14 into 8 and missing part	**Paper Setup / Inner Step:**	**Final Answer:**
7	**Part 6 and Part 7 → Whole?** Strategy Hint: Combine consecutive parts (6 + 7)	**Paper Setup / Inner Step:**	**Final Answer:**
8	**Whole 18, Part 9 → Missing Part?** Strategy Hint: Half of 18 (double fact)	**Paper Setup / Inner Step:**	**Final Answer:**
9	**Part 12 and Part 8 → Whole?** Strategy Hint: Ones digits 2 + 8 make a ten	**Paper Setup / Inner Step:**	**Final Answer:**
10	**Whole 25, Part 11 → Missing Part?** Strategy Hint: Subtract tens and ones (25 - 11)	**Paper Setup / Inner Step:**	**Final Answer:**

Foundational Fluency Check (No-Penalty Primary Scoring)

Award +5 points for each correct answer (0 points for skipped or incorrect items—no negative penalty). Build or solve mentally/on paper first, then check with your partner or teacher on the SC-6!

Unit 3 Fluency Check: Complements to 10 & Make-10 Bridging

Complements to 10, Make-10 Bridging & Place-Value Regrouping • Focus: Place Value Studio (Digit Blocks & Gold [+1] Carry Tokens)

MAX STACK DEPTH: 2 | **TIME LIMIT: 5 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: _____:_____ **Score: _____ / 50**

1	**Complement of 7 to make 10 (7 + ? = 10)** Strategy Hint: How many more from 7 to reach 10?	**Paper Setup / Inner Step:**	**Final Answer:**
2	**Complement of 4 to make 10 (4 + ? = 10)** Strategy Hint: How many more from 4 to reach 10?	**Paper Setup / Inner Step:**	**Final Answer:**
3	**Make-10 Bridge: 8 + 5 = (8 + 2) + 3** Strategy Hint: Complete 10 first, then add remaining 3	**Paper Setup / Inner Step:**	**Final Answer:**
4	**Make-10 Bridge: 9 + 6 = (9 + 1) + 5** Strategy Hint: Complete 10 first, then add remaining 5	**Paper Setup / Inner Step:**	**Final Answer:**
5	**Make-10 Bridge: 7 + 6 = (7 + 3) + 3** Strategy Hint: Complete 10 first, then add remaining 3	**Paper Setup / Inner Step:**	**Final Answer:**
6	**Complement of 1 to make 10 (1 + ? = 10)** Strategy Hint: Ten minus 1	**Paper Setup / Inner Step:**	**Final Answer:**
7	**Subtract from two tens: 20 - 7** Strategy Hint: Decompose 20 into 10 + 10, subtract 7	**Paper Setup / Inner Step:**	**Final Answer:**
8	**Make-10 Bridge: 8 + 7 = (8 + 2) + 5** Strategy Hint: Complete 10 first, then add remaining 5	**Paper Setup / Inner Step:**	**Final Answer:**
9	**1 ten and 6 ones: 10 + 6** Strategy Hint: Place value composition	**Paper Setup / Inner Step:**	**Final Answer:**
10	**Complement of 2 to make 10 (10 - 2)** Strategy Hint: Partner of 2 that makes 10	**Paper Setup / Inner Step:**	**Final Answer:**

Foundational Fluency Check (No-Penalty Primary Scoring)

Award +5 points for each correct answer (0 points for skipped or incorrect items—no negative penalty). Build or solve mentally/on paper first, then check with your partner or teacher on the SC-6!

Unit 4 Fluency Check: Doubles, Near-Doubles & Equal Groups

Twin-Row Doubles, Near-Doubles (+1) & Early Multiplication • Focus: Place Value Studio (Two-Row Tray & Digit Blocks)

MAX STACK DEPTH: 2 | **TIME LIMIT: 5 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: _____:_____ **Score: _____ / 50**

1	**Double 6 (6 + 6)** Strategy Hint: Two equal rows of 6	**Paper Setup / Inner Step:**	**Final Answer:**
2	**Double 8 (8 + 8)** Strategy Hint: Two equal rows of 8	**Paper Setup / Inner Step:**	**Final Answer:**
3	**Near-Double: 7 + 8** Strategy Hint: Double 7, then add 1 more	**Paper Setup / Inner Step:**	**Final Answer:**
4	**Near-Double: 6 + 7** Strategy Hint: Double 6, then add 1 more	**Paper Setup / Inner Step:**	**Final Answer:**
5	**Double 9 (9 + 9)** Strategy Hint: Two equal rows of 9	**Paper Setup / Inner Step:**	**Final Answer:**
6	**Near-Double: 8 + 9** Strategy Hint: Double 8, then add 1 more	**Paper Setup / Inner Step:**	**Final Answer:**
7	**Equal groups: 3 groups of 4 (3 × 4)** Strategy Hint: Three groups of 4 items	**Paper Setup / Inner Step:**	**Final Answer:**
8	**Equal groups: 4 groups of 5 (4 × 5)** Strategy Hint: Four groups of 5 items	**Paper Setup / Inner Step:**	**Final Answer:**
9	**Double 12 (12 + 12)** Strategy Hint: Double the ten and double the ones	**Paper Setup / Inner Step:**	**Final Answer:**
10	**Near-Double: 9 + 10** Strategy Hint: Double 9, then add 1 more	**Paper Setup / Inner Step:**	**Final Answer:**

Foundational Fluency Check (No-Penalty Primary Scoring)

Award +5 points for each correct answer (0 points for skipped or incorrect items—no negative penalty). Build or solve mentally/on paper first, then check with your partner or teacher on the SC-6!

Cumulative Check C1: Primary Math Foundations (Units 1–4)

Comprehensive Primary Fluency Review across Counting, Bonds, Make-10 & Doubles • Focus: Place Value Studio & SC-6 Stack Workbench

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MAX STACK DEPTH: 2 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: _____:_____ **Score: _____ / 50**

1	**Start at 1, add 6 unit steps: 1 + 6** Strategy Hint: Count on 6 steps from 1	**Paper Setup / Inner Step:**	**Final Answer:**
2	**Number bond: Part 9 and Part 7 → Whole?** Strategy Hint: Make 10 (9 + 1) plus 6	**Paper Setup / Inner Step:**	**Final Answer:**
3	**Decompose: Whole 17, Part 8 → Missing Part?** Strategy Hint: Subtract 8 from 17	**Paper Setup / Inner Step:**	**Final Answer:**
4	**Complement of 3 to make 10 (3 + ? = 10)** Strategy Hint: Partner of 3 that makes 10	**Paper Setup / Inner Step:**	**Final Answer:**
5	**Double 14 (14 + 14)** Strategy Hint: Double 10 plus double 4	**Paper Setup / Inner Step:**	**Final Answer:**
6	**Near-double: 11 + 12** Strategy Hint: Double 11, then add 1 more	**Paper Setup / Inner Step:**	**Final Answer:**
7	**Make-10 Bridge: 9 + 8** Strategy Hint: Bridge through 10: (9 + 1) + 7	**Paper Setup / Inner Step:**	**Final Answer:**
8	**Equal groups: 6 groups of 5 (6 × 5)** Strategy Hint: Six groups of 5	**Paper Setup / Inner Step:**	**Final Answer:**
9	**Decompose: Whole 30, Part 12 → Missing Part?** Strategy Hint: Subtract 12 from 30	**Paper Setup / Inner Step:**	**Final Answer:**
10	**Two-step: Double 8, then add 5** Strategy Hint: Compute 8 + 8 first, then add 5	**Paper Setup / Inner Step:**	**Final Answer:**

Foundational Fluency Check (No-Penalty Primary Scoring)

Award +5 points for each correct answer (0 points for skipped or incorrect items—no negative penalty). Build or solve mentally/on paper first, then check with your partner or teacher on the SC-6!

Unit 5 Time Trial: Multi-Step Operations & Inner Groups

Inside-Out Evaluation of Two-Step & Three-Step Arithmetic • Focus: Place Value Studio & 4-Register Stack

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MAX STACK DEPTH: 4 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: _____:_____ **Score: _____ / 50**

1	**(18 - 6) × 3** Strategy Hint: Solve inner difference first, then multiply	**Paper Setup / Inner Step:**	**Final Answer:**
2	**(24 + 16) ÷ 8** Strategy Hint: Solve inner sum first, then divide by 8	**Paper Setup / Inner Step:**	**Final Answer:**
3	**45 ÷ (12 - 3)** Strategy Hint: Evaluate divisor (12 - 3) before dividing 45	**Paper Setup / Inner Step:**	**Final Answer:**
4	**(15 × 4) + (18 ÷ 2)** Strategy Hint: Evaluate both groups, then add	**Paper Setup / Inner Step:**	**Final Answer:**
5	**(50 - 14) ÷ (3 + 6)** Strategy Hint: Find numerator and denominator first	**Paper Setup / Inner Step:**	**Final Answer:**
6	**(12 × 5) - (8 × 4)** Strategy Hint: Compute both products before subtracting	**Paper Setup / Inner Step:**	**Final Answer:**
7	**((20 - 8) × 4) ÷ 6** Strategy Hint: Work from innermost parentheses outward	**Paper Setup / Inner Step:**	**Final Answer:**
8	**(72 ÷ (4 + 5)) × 3** Strategy Hint: Solve (4 + 5) first, divide 72, then × 3	**Paper Setup / Inner Step:**	**Final Answer:**
9	**100 - (16 × 5 - 12)** Strategy Hint: Evaluate inner product and difference first	**Paper Setup / Inner Step:**	**Final Answer:**
10	**((8 + 4) × (9 - 3)) ÷ 12** Strategy Hint: Multiply the two inner groups, then ÷ 12	**Paper Setup / Inner Step:**	**Final Answer:**

UIL Mathematics & Calculator Sprint Scoring

Score (+5 × Correct) − (2 × Incorrect). Skipped items receive 0. Write your inner step on paper and verify within Max Stack Depth 4. Round non-exact decimals to 4 significant digits.

Unit 6 Time Trial: Fractions, Mixed Numbers & Reciprocals

Common Denominators, Fraction Equivalence & Decimal Values • Focus: Fraction Studio (84 mm Towers & Comparator Lid)

MAX STACK DEPTH: 4 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: _____:_____ **Score: _____ / 50**

1	**1/2 + 1/4 (as a decimal)** Strategy Hint: Convert 1/2 to 2/4 and add 1/4	**Paper Setup / Inner Step:**	**Final Answer:**
2	**3/4 - 1/2 (as a decimal)** Strategy Hint: Subtract 2/4 from 3/4	**Paper Setup / Inner Step:**	**Final Answer:**
3	**2 1/2 + 1 1/4 (as a decimal)** Strategy Hint: Add whole parts and fourths	**Paper Setup / Inner Step:**	**Final Answer:**
4	**Reciprocal of 4/5 (1 ÷ 4/5 as a decimal)** Strategy Hint: Invert numerator and denominator: 5/4	**Paper Setup / Inner Step:**	**Final Answer:**
5	**3/8 + 1/4 (as a decimal)** Strategy Hint: Rewrite 1/4 as 2/8, then add to 3/8	**Paper Setup / Inner Step:**	**Final Answer:**
6	**2/3 × 3/4 (as a decimal)** Strategy Hint: Cancel common factor 3: 2/4 = 1/2	**Paper Setup / Inner Step:**	**Final Answer:**
7	**3/4 ÷ 1/2 (as a decimal)** Strategy Hint: How many halves fit in 3/4? (3/4 × 2)	**Paper Setup / Inner Step:**	**Final Answer:**
8	**1 1/2 × 4** Strategy Hint: Multiply (1 × 4) + (1/2 × 4)	**Paper Setup / Inner Step:**	**Final Answer:**
9	**5 - 2 3/4 (as a decimal)** Strategy Hint: Regroup 5 as 4 4/4 and subtract 2 3/4	**Paper Setup / Inner Step:**	**Final Answer:**
10	**(1/3 + 1/6) × 12** Strategy Hint: Combine 2/6 + 1/6 = 1/2 first, then × 12	**Paper Setup / Inner Step:**	**Final Answer:**

UIL Mathematics & Calculator Sprint Scoring

Score (+5 × Correct) − (2 × Incorrect). Skipped items receive 0. Write your inner step on paper and verify within Max Stack Depth 4. Round non-exact decimals to 4 significant digits.

Unit 7 Time Trial: Nested Expressions & Chained Branches

Inside-Out Evaluation of Nested Fractions & Multi-Branch Trees • Focus: Prime Factor Studio & 4-Register Stack

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MAX STACK DEPTH: 4 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: _____:_____ **Score: _____ / 50**

1	$(8 + 2) \times (9 - 4)$ Strategy Hint: Solve both parenthesized groups, then ×	**Paper Setup / Inner Step:**	**Final Answer:**
2	$((14 - 6) \times 3) + (24 \div 4)$ Strategy Hint: Evaluate left and right branches, then +	**Paper Setup / Inner Step:**	**Final Answer:**
3	$(30 - (4 \times 5)) \times (12 \div (2 + 4))$ Strategy Hint: Start with innermost (4×5) and (2+4)	**Paper Setup / Inner Step:**	**Final Answer:**
4	$((5 + 3) \times (8 - 2)) \div (6 - 2)$ Strategy Hint: Evaluate all three parenthesized groups	**Paper Setup / Inner Step:**	**Final Answer:**
5	$((18 \div 3) + 4) \times ((15 - 7) \div 2)$ Strategy Hint: Resolve inner quotient and difference first	**Paper Setup / Inner Step:**	**Final Answer:**
6	$(10 + 5 \times 2) \div (12 - 2 \times 4)$ Strategy Hint: Multiply before adding/subtracting	**Paper Setup / Inner Step:**	**Final Answer:**
7	$(25 - 5 \times 3) \times (4 + 6 \div 2)$ Strategy Hint: Evaluate inner product and quotient first	**Paper Setup / Inner Step:**	**Final Answer:**
8	$((7 + 3) \times 4 - 8) \div 4$ Strategy Hint: Work outward from (7 + 3)	**Paper Setup / Inner Step:**	**Final Answer:**
9	$(48 \div (2 + 6)) \times (15 - 3 \times 3)$ Strategy Hint: Evaluate (2 + 6) and (3 × 3) first	**Paper Setup / Inner Step:**	**Final Answer:**
10	$(16 - (4 + 2)) \times (18 \div (3 \times 2))$ Strategy Hint: Resolve innermost sum and product first	**Paper Setup / Inner Step:**	**Final Answer:**

UIL Mathematics & Calculator Sprint Scoring

Score (+5 × Correct) − (2 × Incorrect). Skipped items receive 0. Write your inner step on paper and verify within Max Stack Depth 4. Round non-exact decimals to 4 significant digits.

Unit 8 Time Trial: Polynomials & Quadratic Factoring

Prime Factor Pairs, Signed Trinomials & Quadratic Roots • Focus: Place Value Studio & Prime Factor Studio

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MAX STACK DEPTH: 4 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: _____:_____ **Score: _____ / 50**

1	**Greater root of $x^2 + 5x + 6 = 0$** Strategy Hint: Factor 6 into p × q = 6 with p + q = 5	**Paper Setup / Inner Step:**	**Final Answer:**
2	**Greater root of $x^2 + 7x + 12 = 0$** Strategy Hint: Factor 12 into p × q = 12 with p + q = 7	**Paper Setup / Inner Step:**	**Final Answer:**
3	**Greater root of $x^2 + 8x + 15 = 0$** Strategy Hint: Factor 15 into p × q = 15 with p + q = 8	**Paper Setup / Inner Step:**	**Final Answer:**
4	**Positive root of $x^2 - x - 6 = 0$** Strategy Hint: Opposite-sign factors of -6 summing to -1	**Paper Setup / Inner Step:**	**Final Answer:**
5	**Lesser root of $x^2 - 5x + 6 = 0$** Strategy Hint: Two negative factors of +6 summing to -5	**Paper Setup / Inner Step:**	**Final Answer:**
6	**Positive root of $x^2 - 9 = 0$** Strategy Hint: Difference of squares: (x - a)(x + a)	**Paper Setup / Inner Step:**	**Final Answer:**
7	**Positive root of $x^2 - 16 = 0$** Strategy Hint: Difference of squares: $x^2 = 16$	**Paper Setup / Inner Step:**	**Final Answer:**
8	**Evaluate $2x^3 + 4x - 5$ at $x = 3$** Strategy Hint: Substitute x = 3 into each term	**Paper Setup / Inner Step:**	**Final Answer:**
9	**Greater root of $2x^2 + 5x + 2 = 0$** Strategy Hint: Factor (2x + 1)(x + 2) = 0	**Paper Setup / Inner Step:**	**Final Answer:**
10	**Repeated root of $x^2 + 10x + 25 = 0$** Strategy Hint: Perfect square trinomial: $(x + p)^2 = 0$	**Paper Setup / Inner Step:**	**Final Answer:**

UIL Mathematics & Calculator Sprint Scoring

Score (+5 × Correct) − (2 × Incorrect). Skipped items receive 0. Write your inner step on paper and verify within Max Stack Depth 4. Round non-exact decimals to 4 significant digits.

Cumulative Trial C2: Intermediate Mathematics Championship

Comprehensive Time Trial across Units 1–8 (Fractions, Expressions & Algebra) • Focus: Place Value, Fraction & Prime Factor Studios

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MAX STACK DEPTH: 4 | **TIME LIMIT: 15 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: _____:_____ **Score: _____ / 50**

1	**((32 - 14) × 5) ÷ 6** Strategy Hint: Evaluate inner difference first	**Paper Setup / Inner Step:**	**Final Answer:**
2	**3/4 + (1/2 × 1/4) as a decimal** Strategy Hint: Multiply 1/2 × 1/4 before adding 3/4	**Paper Setup / Inner Step:**	**Final Answer:**
3	**Distance between P1(3, 4) and P2(9, 12)** Strategy Hint: Pythagorean distance $\sqrt{(dx^2 + dy^2)}$	**Paper Setup / Inner Step:**	**Final Answer:**
4	**3 1/2 - 1 3/4 (as a decimal)** Strategy Hint: Regroup 3 2/4 as 2 6/4 to subtract 1 3/4	**Paper Setup / Inner Step:**	**Final Answer:**
5	**((15 + 5) × (8 - 3)) ÷ 10** Strategy Hint: Multiply the two inner branches, then ÷ 10	**Paper Setup / Inner Step:**	**Final Answer:**
6	**Double 18, then subtract 9** Strategy Hint: Compute 18 + 18 first, then subtract 9	**Paper Setup / Inner Step:**	**Final Answer:**
7	**Slope between P1(-1, -2) and P2(3, 10)** Strategy Hint: Slope m = (y2 - y1) / (x2 - x1)	**Paper Setup / Inner Step:**	**Final Answer:**
8	**80 - (15 × 4) + 12** Strategy Hint: Compute inner product 15 × 4 first	**Paper Setup / Inner Step:**	**Final Answer:**
9	**1 ÷ (1/4 + 1/6) as a decimal** Strategy Hint: Add 3/12 + 2/12 first, then take reciprocal	**Paper Setup / Inner Step:**	**Final Answer:**
10	**Distance from origin (0, 0) to (12, 16)** Strategy Hint: Hypotenuse $\sqrt{(12^2 + 16^2)}$	**Paper Setup / Inner Step:**	**Final Answer:**

UIL Mathematics & Calculator Sprint Scoring

Score (+5 × Correct) − (2 × Incorrect). Skipped items receive 0. Write your inner step on paper and verify within Max Stack Depth 4. Round non-exact decimals to 4 significant digits.

Unit 9 Time Trial: Trigonometry, Vectors & Polar Coordinates

Right Triangles, Polar Conversions & Vector Decomposition • Focus: Fraction Studio (84 mm Ratio Towers) & Place Value Studio

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MAX STACK DEPTH: 4 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: _____:_____ **Score: _____ / 50**

1 **Evaluate sin(30°)**

Strategy Hint: Opposite / Hypotenuse in a 30°-60°-90° triangle

Paper Setup / Inner Step: _____ Final Answer: _____

2 **Evaluate cos(60°)**

Strategy Hint: Adjacent / Hypotenuse at 60°

Paper Setup / Inner Step: _____ Final Answer: _____

3 **Evaluate tan(45°)**

Strategy Hint: Opposite equals Adjacent at 45°

Paper Setup / Inner Step: _____ Final Answer: _____

4 **Hypotenuse = 10, $\vartheta = 30°$ → Opposite leg y**

Strategy Hint: Opposite leg y = r × sin(ϑ)

Paper Setup / Inner Step: _____ Final Answer: _____

5 **Hypotenuse = 12, $\vartheta = 60°$ → Adjacent leg x**

Strategy Hint: Adjacent leg x = r × cos(ϑ)

Paper Setup / Inner Step: _____ Final Answer: _____

6 **Polar ($r = 10$, $\vartheta = 0°$) → Horizontal component x**

Strategy Hint: Horizontal projection x = r × cos(0°)

Paper Setup / Inner Step: _____ Final Answer: _____

7 **Vector <3, 4> → Magnitude r**

Strategy Hint: Pythagorean magnitude $\sqrt{(x^2 + y^2)}$

Paper Setup / Inner Step: _____ Final Answer: _____

8 **Vector <5, 12> → Magnitude r**

Strategy Hint: Pythagorean magnitude $\sqrt{(5^2 + 12^2)}$

Paper Setup / Inner Step: _____ Final Answer: _____

9 **Vector <8, 6> → Magnitude r**

Strategy Hint: Pythagorean magnitude $\sqrt{(8^2 + 6^2)}$

Paper Setup / Inner Step: _____ Final Answer: _____

10 **Solve sin(ϑ) = 0.5 for acute angle ϑ (in degrees)**

Strategy Hint: Inverse sine arcsin(0.5)

Paper Setup / Inner Step: _____ Final Answer: _____

UIL Mathematics & Calculator Sprint Scoring

Score (+5 × Correct) − (2 × Incorrect). Skipped items receive 0. Write your inner step on paper and verify within Max Stack Depth 4. Round non-exact decimals to 4 significant digits.

TIMED SPEED & FLUENCY SPRINT • SC-32 SCIENTIFIC

Unit 10 Time Trial: Exponential Growth & Logarithms

Natural Base e, Continuous Compounding & Doubling Time • Focus: Prime Factor Studio & Fraction Studio Towers

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MAX STACK DEPTH: 4 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: _____:_____ **Score: _____ / 50**

1 **Evaluate 2^5**

Strategy Hint: Multiply 2 by itself 5 times

Paper Setup / Inner Step:

Final Answer:

2 **Evaluate 3^4**

Strategy Hint: Multiply 3 by itself 4 times

Paper Setup / Inner Step:

Final Answer:

3 **Evaluate ln(e^1)**

Strategy Hint: Natural logarithm is the inverse of e^x

Paper Setup / Inner Step:

Final Answer:

4 **Evaluate e^0**

Strategy Hint: Any nonzero base to the 0 power

Paper Setup / Inner Step:

Final Answer:

5 **Evaluate log10(1000)**

Strategy Hint: What power of 10 equals 1,000?

Paper Setup / Inner Step:

Final Answer:

6 **Compound interest: 1000 × (1.05)^2**

Strategy Hint: Square growth factor 1.05 first, then × 1000

Paper Setup / Inner Step:

Final Answer:

7 **Continuous doubling time: ln(2) ÷ 0.07**

Strategy Hint: Divide ln(2) by continuous rate r = 0.07

Paper Setup / Inner Step:

Final Answer:

8 **Half-life decay after 3 periods: 100 × (0.5)^3**

Strategy Hint: Cube 0.5 first, then multiply by 100

Paper Setup / Inner Step:

Final Answer:

9 **Continuous growth: 500 × e^(0.04 × 5)**

Strategy Hint: Evaluate inner exponent r×t first, then e^x

Paper Setup / Inner Step:

Final Answer:

10 **Evaluate 2^10**

Strategy Hint: Ten doublings (1 kilobyte)

Paper Setup / Inner Step:

Final Answer:

UIL Mathematics & Calculator Sprint Scoring

Score (+5 × Correct) − (2 × Incorrect). Skipped items receive 0. Write your inner step on paper and verify within Max Stack Depth 4. Round non-exact decimals to 4 significant digits.

TIMED SPEED & FLUENCY SPRINT • SC-32 SCIENTIFIC

Unit 11 Time Trial: Combinatorics, Statistics & Probability

Factorials, Permutations, Combinations & Sample Summations • Focus: Fraction Studio (Probability Ratio Towers) & Prime Factor Studio

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MAX STACK DEPTH: 4 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: _____:_____ **Score: _____ / 50**

1	**Evaluate 5! (5 factorial)** Strategy Hint: Product 5 × 4 × 3 × 2 × 1	Paper Setup / Inner Step:	Final Answer:
2	**Evaluate 6! (6 factorial)** Strategy Hint: Product 6 × 5!	Paper Setup / Inner Step:	Final Answer:
3	**Permutations 5P2 = 5! ÷ 3!** Strategy Hint: Cancel 3! to leave 5 × 4	Paper Setup / Inner Step:	Final Answer:
4	**Permutations 6P3 = 6! ÷ 3!** Strategy Hint: Cancel 3! to leave 6 × 5 × 4	Paper Setup / Inner Step:	Final Answer:
5	**Combinations 5C2 = 5! ÷ (2! × 3!)** Strategy Hint: Unordered pairs chosen from 5	Paper Setup / Inner Step:	Final Answer:
6	**Combinations 6C2 = 6! ÷ (2! × 4!)** Strategy Hint: Unordered pairs chosen from 6	Paper Setup / Inner Step:	Final Answer:
7	**Combinations 7C3 = 7! ÷ (3! × 4!)** Strategy Hint: Unordered triples chosen from 7	Paper Setup / Inner Step:	Final Answer:
8	**Sum of first 5 positive integers: 1 + 2 + 3 + 4 + 5** Strategy Hint: Triangular number n(n + 1)/2	Paper Setup / Inner Step:	Final Answer:
9	**Evaluate 4! (4 factorial)** Strategy Hint: Product 4 × 3 × 2 × 1	Paper Setup / Inner Step:	Final Answer:
10	**Probability of 1 specific pair out of 6C2: 1 ÷ 6C2** Strategy Hint: Reciprocal of total combinations 6C2	Paper Setup / Inner Step:	Final Answer:

UIL Mathematics & Calculator Sprint Scoring

Score (+5 × Correct) − (2 × Incorrect). Skipped items receive 0. Write your inner step on paper and verify within Max Stack Depth 4. Round non-exact decimals to 4 significant digits.

Unit 12 Time Trial: Digital Logic, Base Conversions & CS

Binary & Hexadecimal Place Value and Bitwise Boolean Logic • Focus: Place Value Studio (Binary [0]/[1] Register Trays)

MAX STACK DEPTH: 4 | **TIME LIMIT: 10 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: _____:_____ **Score: _____ / 50**

1 **Convert decimal 255 to hexadecimal (Base 16)**

Strategy Hint: Express 255 as $15 \times 16^1 + 15 \times 16^0$

Paper Setup / Inner Step: _____ Final Answer: _____

2 **Convert hexadecimal 10_16 to decimal (Base 10)**

Strategy Hint: Place value: $1 \times 16^1 + 0 \times 16^0$

Paper Setup / Inner Step: _____ Final Answer: _____

3 **Convert decimal 15 to 4-bit binary (Base 2)**

Strategy Hint: Sum of powers of 2: $8 + 4 + 2 + 1$

Paper Setup / Inner Step: _____ Final Answer: _____

4 **Convert binary 1010_2 to decimal (Base 10)**

Strategy Hint: Place value: $8 + 0 + 2 + 0$

Paper Setup / Inner Step: _____ Final Answer: _____

5 **Bitwise AND: 12 (1100_2) AND 10 (1010_2) in decimal**

Strategy Hint: Compare 4-bit 1100 and 1010 bit-by-bit

Paper Setup / Inner Step: _____ Final Answer: _____

6 **Bitwise OR: 12 (1100_2) OR 10 (1010_2) in decimal**

Strategy Hint: Bit is 1 if either input bit is 1

Paper Setup / Inner Step: _____ Final Answer: _____

7 **Bitwise XOR: 12 (1100_2) XOR 10 (1010_2) in decimal**

Strategy Hint: Bit is 1 only when input bits differ

Paper Setup / Inner Step: _____ Final Answer: _____

8 **Convert hexadecimal 20_16 to decimal (Base 10)**

Strategy Hint: Place value: $2 \times 16^1 + 0 \times 16^0$

Paper Setup / Inner Step: _____ Final Answer: _____

9 **Convert decimal 64 to octal (Base 8)**

Strategy Hint: Express 64 in powers of 8 (8^2)

Paper Setup / Inner Step: _____ Final Answer: _____

10 **Bitwise AND: 255 (11111111_2) AND 15 (00001111_2)**

Strategy Hint: Masking the low 4 bits

Paper Setup / Inner Step: _____ Final Answer: _____

UIL Mathematics & Calculator Sprint Scoring

Score (+5 × Correct) − (2 × Incorrect). Skipped items receive 0. Write your inner step on paper and verify within Max Stack Depth 4. Round non-exact decimals to 4 significant digits.

Cumulative Trial C3: Grand Mathematics & Stack Invitational

Master UIL-Style Time Trial across Units 1–12 (Grades 7–12+) • Focus: All-Level StackCalc Mathematics Lab (Place Value, Fraction & Prime Studios)

MAX STACK DEPTH: 4 | **TIME LIMIT: 20 MIN** | **MAX SCORE: 50 PTS**

Competitor: _____ Date: _____ Elapsed: _____:_____ **Score: _____ / 50**

1	**Monthly compounding: 250 × (1 + 0.06 ÷ 12)^24** Strategy Hint: Evaluate inner periodic rate first	**Paper Setup / Inner Step:**	**Final Answer:**
2	**Distance between P1(-3, 5) and P2(5, -10)** Strategy Hint: Pythagorean distance $\sqrt{(dx^2 + dy^2)}$	**Paper Setup / Inner Step:**	**Final Answer:**
3	**Evaluate 12 cos(45°) + 5 sin(45°)** Strategy Hint: Sum the two trigonometric projections	**Paper Setup / Inner Step:**	**Final Answer:**
4	**(3 1/3 ÷ 1 1/4) + 2/5 (to 4 decimal places)** Strategy Hint: Divide mixed numbers first, then add 2/5	**Paper Setup / Inner Step:**	**Final Answer:**
5	**Combinations 8C3 = 8! ÷ (3! × 5!)** Strategy Hint: Cancel 5! to leave (8 × 7 × 6) ÷ 6	**Paper Setup / Inner Step:**	**Final Answer:**
6	**Continuous growth: 1500 × e^(0.05 × 10)** Strategy Hint: Compute exponent 0.05 × 10 first	**Paper Setup / Inner Step:**	**Final Answer:**
7	**((35 - 7) × 4) ÷ (14 + 2)** Strategy Hint: Evaluate numerator and denominator first	**Paper Setup / Inner Step:**	**Final Answer:**
8	**Subnet mask: 192 (11000000_2) AND 240 (11110000_2)** Strategy Hint: Compare top 4 bits of 192 and 240	**Paper Setup / Inner Step:**	**Final Answer:**
9	**Sum of double 6 and near-double (8 + 9)** Strategy Hint: Compute (6 + 6) + (8 + 9)	**Paper Setup / Inner Step:**	**Final Answer:**
10	**Convert decimal 4095 (2^12 - 1) to hexadecimal** Strategy Hint: Twelve 1-bits grouped into three hex nibbles	**Paper Setup / Inner Step:**	**Final Answer:**

UIL Mathematics & Calculator Sprint Scoring

Score (+5 × Correct) − (2 × Incorrect). Skipped items receive 0. Write your inner step on paper and verify within Max Stack Depth 4. Round non-exact decimals to 4 significant digits.

Official Solutions & Keystrokes — Part 1

Verified answers and calculator check sequences for Part 1 (Trials 1–6 & Cumulative C1).

Unit 1 Fluency Check: Step Counting & One-More Addition (Stack 1)

1. Ans: 3 — [1] [ENTER] [1] [+] [1] [+]
2. Ans: 7 — [5] [ENTER] [1] [+] [1] [+]
3. Ans: 4 — [1] [ENTER] [+] [+] [+]
4. Ans: 8 — [1] [ENTER] [+] [+] [+] [+] [+] [+] [+]
5. Ans: 4 — [6] [ENTER] [1] [-] [1] [-]
6. Ans: 5 — [C] [1] [ENTER] [+] [+] [+] [+]
7. Ans: 6 — [2] [ENTER] [2] [+] [2] [+]
8. Ans: 10 — [7] [ENTER] [1] [+] [1] [+] [1] [+]
9. Ans: 9 — [0] [ENTER] [9] [+]
10. Ans: 7 — [10] [ENTER] [1] [-] [1] [-] [1] [-]

Unit 2 Fluency Check: Part-Part-Whole Number Bonds (Stack 2)

1. Ans: 7 — [4] [ENTER] [3] [+]
2. Ans: 10 — [5] [ENTER] [5] [+]
3. Ans: 3 — [9] [ENTER] [6] [-]
4. Ans: 6 — [10] [ENTER] [4] [-]
5. Ans: 15 — [8] [ENTER] [7] [+]
6. Ans: 6 — [14] [ENTER] [8] [-]
7. Ans: 13 — [6] [ENTER] [7] [+]
8. Ans: 9 — [18] [ENTER] [9] [-]
9. Ans: 20 — [12] [ENTER] [8] [+]
10. Ans: 14 — [25] [ENTER] [11] [-]

Unit 3 Fluency Check: Complements to 10 & Make-10 Bridging (Stack 2)

1. Ans: 3 — [10] [ENTER] [7] [-]
2. Ans: 6 — [10] [ENTER] [4] [-]
3. Ans: 13 — [8] [ENTER] [2] [+] [3] [+]
4. Ans: 15 — [9] [ENTER] [1] [+] [5] [+]
5. Ans: 13 — [7] [ENTER] [3] [+] [3] [+]
6. Ans: 9 — [10] [ENTER] [1] [-]
7. Ans: 13 — [20] [ENTER] [7] [-]
8. Ans: 15 — [8] [ENTER] [2] [+] [5] [+]
9. Ans: 16 — [10] [ENTER] [6] [+]
10. Ans: 8 — [10] [ENTER] [2] [-]

Unit 4 Fluency Check: Doubles, Near-Doubles & Equal Groups (Stack 2)

1. Ans: 12 — [6] [ENTER] [+]
2. Ans: 16 — [8] [ENTER] [+]
3. Ans: 15 — [7] [ENTER] [+] [1] [+]
4. Ans: 13 — [6] [ENTER] [+] [1] [+]
5. Ans: 18 — [9] [ENTER] [+]
6. Ans: 17 — [8] [ENTER] [+] [1] [+]
7. Ans: 12 — [4] [ENTER] [3] [×]
8. Ans: 20 — [5] [ENTER] [4] [×]
9. Ans: 24 — [12] [ENTER] [+]
10. Ans: 19 — [9] [ENTER] [+] [1] [+]

Cumulative Check C1: Primary Math Foundations (Units 1–4) (Stack 2)

1. Ans: 7 — [1] [ENTER] [+] [+] [+] [+] [+] [+]
2. Ans: 16 — [9] [ENTER] [7] [+]
3. Ans: 9 — [17] [ENTER] [8] [-]
4. Ans: 7 — [10] [ENTER] [3] [-]
5. Ans: 28 — [14] [ENTER] [+]
6. Ans: 23 — [11] [ENTER] [+] [1] [+]
7. Ans: 17 — [9] [ENTER] [1] [+] [7] [+]
8. Ans: 30 — [5] [ENTER] [6] [×]
9. Ans: 18 — [30] [ENTER] [12] [-]
10. Ans: 21 — [8] [ENTER] [+] [5] [+]

Unit 5 Time Trial: Multi-Step Operations & Inner Groups (Stack 4)

1. Ans: 36 — [18] [ENTER] [6] [-] [3] [×]
2. Ans: 5 — [24] [ENTER] [16] [+] [8] [÷]
3. Ans: 5 — [45] [ENTER] [12] [ENTER] [3] [-] [÷]
4. Ans: 69 — [15] [ENTER] [4] [×] [18] [ENTER] [2] [÷] [+]
5. Ans: 4 — [50] [ENTER] [14] [-] [3] [ENTER] [6] [+] [÷]
6. Ans: 28 — [12] [ENTER] [5] [×] [8] [ENTER] [4] [×] [-]
7. Ans: 8 — [20] [ENTER] [8] [-] [4] [×] [6] [÷]
8. Ans: 24 — [72] [ENTER] [4] [ENTER] [5] [+] [÷] [3] [×]
9. Ans: 32 — [100] [ENTER] [16] [ENTER] [5] [×] [12] [-] [-]
10. Ans: 6 — [8] [ENTER] [4] [+] [9] [ENTER] [3] [-] [×] [12] [÷]

Unit 6 Time Trial: Fractions, Mixed Numbers & Reciprocals (Stack 4)

1. Ans: 0.75 — [.1.2] [ENTER] [.1.4] [+]
2. Ans: 0.25 — [.3.4] [ENTER] [.1.2] [-]
3. Ans: 3.75 — [2.1.2] [ENTER] [1.1.4] [+]
4. Ans: 1.25 — [.4.5] [1/x]
5. Ans: 0.625 — [.3.8] [ENTER] [.1.4] [+]
6. Ans: 0.5 — [.2.3] [ENTER] [.3.4] [×]
7. Ans: 1.5 — [.3.4] [ENTER] [.1.2] [÷]
8. Ans: 6 — [1.1.2] [ENTER] [4] [×]
9. Ans: 2.25 — [5] [ENTER] [2.3.4] [-]
10. Ans: 6 — [.1.3] [ENTER] [.1.6] [+] [12] [×]

Scoring Adjudication

Accept equivalent rational fractions or 4-sig-fig decimals (e.g., 3/4 or 0.75). Primary Checks (Trials 1–4, C1) award +5 per correct item with no deduction for errors.

Official Solutions & Keystrokes — Part 2

Verified answers and calculator check sequences for Part 2 (Trials 7–12 & Cumulative C2–C3).

Unit 7 Time Trial: Nested Expressions & Chained Branches (Stack 4)

1. Ans: 50 — [8] [ENTER] [2] [+] [9] [ENTER] [4] [-] [×]
2. Ans: 30 — [14] [ENTER] [6] [-] [3] [×] [24] [ENTER] [4] [÷] [+]
3. Ans: 20 — [30] [ENTER] [4] [ENTER] [5] [×] [-] [12] [ENTER] [2] [ENTER] [4] [+] [÷] [×]
4. Ans: 12 — [5] [ENTER] [3] [+] [8] [ENTER] [2] [-] [×] [6] [ENTER] [2] [-] [÷]
5. Ans: 40 — [18] [ENTER] [3] [÷] [4] [+] [15] [ENTER] [7] [-] [2] [÷] [×]
6. Ans: 5 — [10] [ENTER] [5] [ENTER] [2] [×] [+] [12] [ENTER] [2] [ENTER] [4] [×] [-] [÷]
7. Ans: 70 — [25] [ENTER] [5] [ENTER] [3] [×] [-] [4] [ENTER] [6] [ENTER] [2] [÷] [+] [×]
8. Ans: 8 — [7] [ENTER] [3] [+] [4] [×] [8] [-] [4] [÷]
9. Ans: 36 — [48] [ENTER] [2] [ENTER] [6] [+] [÷] [15] [ENTER] [3] [ENTER] [3] [×] [-] [×]
10. Ans: 30 — [16] [ENTER] [4] [ENTER] [2] [+] [-] [18] [ENTER] [3] [ENTER] [2] [×] [÷] [×]

Unit 8 Time Trial: Polynomials & Quadratic Factoring (Stack 4)

1. Ans: -2 — [1] [ENTER] [5] [ENTER] [6] [POLY]
2. Ans: -3 — [1] [ENTER] [7] [ENTER] [12] [POLY]
3. Ans: -3 — [1] [ENTER] [8] [ENTER] [15] [POLY]
4. Ans: 3 — [1] [ENTER] [1] [+/-] [ENTER] [6] [+/-] [POLY]
5. Ans: 2 — [1] [ENTER] [5] [+/-] [ENTER] [6] [POLY]
6. Ans: 3 — [1] [ENTER] [0] [ENTER] [9] [+/-] [POLY]
7. Ans: 4 — [1] [ENTER] [0] [ENTER] [16] [+/-] [POLY]
8. Ans: 25 — [3] [ENTER] [x^2] [2] [×] [3] [ENTER] [4] [×] [+] [5] [-]
9. Ans: -0.5 — [2] [ENTER] [5] [ENTER] [2] [POLY]
10. Ans: -5 — [1] [ENTER] [10] [ENTER] [25] [POLY]

Cumulative Trial C2: Intermediate Mathematics Championship (Stack 4)

1. Ans: 15 — [32] [ENTER] [14] [-] [5] [×] [6] [÷]
2. Ans: 0.875 — [.1.2] [ENTER] [.1.4] [×] [.3.4] [+]
3. Ans: 10 — [9] [ENTER] [3] [-] [x^2] [12] [ENTER] [4] [-] [x^2] [+] [√x]
4. Ans: 1.75 — [3.1.2] [ENTER] [1.3.4] [-]
5. Ans: 10 — [15] [ENTER] [5] [+] [8] [ENTER] [3] [-] [×] [10] [÷]
6. Ans: 27 — [18] [ENTER] [+] [9] [-]
7. Ans: 3 — [10] [ENTER] [2] [+/-] [-] [3] [ENTER] [1] [+/-] [-] [÷]
8. Ans: 32 — [80] [ENTER] [15] [ENTER] [4] [×] [-] [12] [+]
9. Ans: 2.4 — [.1.4] [ENTER] [.1.6] [+] [1/x]
10. Ans: 20 — [12] [x^2] [16] [x^2] [+] [√x]

Unit 9 Time Trial: Trigonometry, Vectors & Polar Coordinates (Stack 4)

1. Ans: 0.5 — [30] [SIN]
2. Ans: 0.5 — [60] [COS]
3. Ans: 1 — [45] [TAN]
4. Ans: 5 — [10] [ENTER] [30] [SIN] [×]
5. Ans: 6 — [12] [ENTER] [60] [COS] [×]
6. Ans: 10 — [10] [ENTER] [0] [yellow] [→y,x]
7. Ans: 5 — [3] [ENTER] [4] [yellow] [→ϑ,r]
8. Ans: 13 — [5] [ENTER] [12] [yellow] [→ϑ,r]
9. Ans: 10 — [8] [ENTER] [6] [yellow] [→ϑ,r]
10. Ans: 30 — [.5] [yellow] [ASIN]

Unit 10 Time Trial: Exponential Growth & Logarithms (Stack 4)

1. Ans: 32 — [2] [ENTER] [5] [y^x]
2. Ans: 81 — [3] [ENTER] [4] [y^x]
3. Ans: 1 — [1] [e^x] [LN]
4. Ans: 1 — [0] [e^x]
5. Ans: 3 — [1000] [yellow] [LOG]
6. Ans: 1102.5 — [1000] [ENTER] [1.05] [ENTER] [2] [y^x] [×]
7. Ans: 9.9021 — [2] [LN] [.07] [÷]
8. Ans: 12.5 — [100] [ENTER] [.5] [ENTER] [3] [y^x] [×]
9. Ans: 610.7014 — [.04] [ENTER] [5] [×] [e^x] [500] [×]
10. Ans: 1024 — [2] [ENTER] [10] [y^x]

Unit 11 Time Trial: Combinatorics, Statistics & Probability (Stack 4)

1. Ans: 120 — [5] [yellow] [x!]
2. Ans: 720 — [6] [yellow] [x!]
3. Ans: 20 — [5] [yellow] [x!] [3] [yellow] [x!] [÷]
4. Ans: 120 — [6] [yellow] [x!] [3] [yellow] [x!] [÷]
5. Ans: 10 — [5] [yellow] [x!] [2] [yellow] [x!] [3] [yellow] [x!] [×] [÷]
6. Ans: 15 — [6] [yellow] [x!] [2] [yellow] [x!] [4] [yellow] [x!] [×] [÷]
7. Ans: 35 — [7] [yellow] [x!] [3] [yellow] [x!] [4] [yellow] [x!] [×] [÷]
8. Ans: 15 — [1] [Σ+] [2] [Σ+] [3] [Σ+] [4] [Σ+] [5] [Σ+]
9. Ans: 24 — [4] [yellow] [x!]
10. Ans: 0.0667 — [6] [yellow] [x!] [2] [yellow] [x!] [4] [yellow] [x!] [×] [÷] [1/x]

Unit 12 Time Trial: Digital Logic, Base Conversions & CS (Stack 4)

1. Ans: FF — [255] [yellow] [BASE] [HEX]
2. Ans: 16 — [HEX] [10] [yellow] [BASE] [DEC]
3. Ans: 1111 — [15] [yellow] [BASE] [BIN]
4. Ans: 10 — [BIN] [1010] [yellow] [BASE] [DEC]
5. Ans: 8 — [12] [ENTER] [10] [AND]
6. Ans: 14 — [12] [ENTER] [10] [OR]
7. Ans: 6 — [12] [ENTER] [10] [XOR]
8. Ans: 32 — [HEX] [20] [yellow] [BASE] [DEC]
9. Ans: 100 — [64] [yellow] [BASE] [OCT]
10. Ans: 15 — [255] [ENTER] [15] [AND]

Cumulative Trial C3: Grand Mathematics & Stack Invitational (Stack 4)

1. Ans: 281.79 — [.06] [ENTER] [12] [÷] [1] [+] [24] [y^x] [250] [×]
2. Ans: 17 — [5] [ENTER] [3] [+/-] [-] [x^2] [10] [+/-] [ENTER] [5] [-] [x^2] [+] [√x]
3. Ans: 12.0208 — [12] [ENTER] [45] [COS] [×] [5] [ENTER] [45] [SIN] [×] [+]
4. Ans: 3.0667 — [3.1.3] [ENTER] [1.1.4] [÷] [.2.5] [+]
5. Ans: 56 — [8] [yellow] [x!] [3] [yellow] [x!] [5] [yellow] [x!] [×] [÷]
6. Ans: 2473.08 — [.05] [ENTER] [10] [×] [e^x] [1500] [×]
7. Ans: 7 — [35] [ENTER] [7] [-] [4] [×] [14] [ENTER] [2] [+] [÷]
8. Ans: 192 — [192] [ENTER] [240] [AND]
9. Ans: 29 — [6] [ENTER] [+] [8] [ENTER] [+] [1] [+] [+]
10. Ans: FFF — [4095] [yellow] [BASE] [HEX]

Scoring Adjudication

Accept equivalent rational fractions or 4-sig-fig decimals (e.g., 3/4 or 0.75). Primary Checks (Trials 1–4, C1) award +5 per correct item with no deduction for errors.

Inside-Out Math & Stack Reference

Five inside-out mathematical patterns and their matching verification sequences on your four-line calculator.

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1. START AT THE DEEPEST INNER EXPRESSION (ZERO PARENTHESES)

Evaluate the most deeply nested quantity first so intermediate results wait in Register Y. Example: For 100 ÷ (3 + 4×5), compute 4×5 + 3 = 23 first, then type 100 and swap:

Keycap Sequence: 

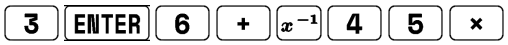
2. INSTANT DOUBLING & SQUARING ([ENTER] [+] AND [ENTER] [×])

Pressing [ENTER] duplicates Register X into Register Y without retyping digits. To double 18 and add 5 in 5 keystrokes:

Keycap Sequence: 

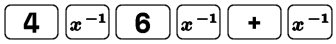
3. NON-COMMUTATIVE SWAP (X SWAP Y) & RECIPROCAL ([1/x])

If you computed the denominator D first and then typed numerator N into X, press [x swap y] [÷]—or invert D in place with [1/x] and multiply:

Keycap Sequence: 

4. PARALLEL RESISTANCE & HARMONIC SUMS (AUTOMATIC STACK LIFT)

To compute 1 / (1/4 + 1/6), never use memory registers—automatic stack lift pushes the first reciprocal into Y while the second is computed:

Keycap Sequence: 

5. VECTOR MAGNITUDE & POLAR CONVERSION (TO POLAR)

To find the hypotenuse or magnitude of components (x=5, y=12) in a single operation on the SC-32:

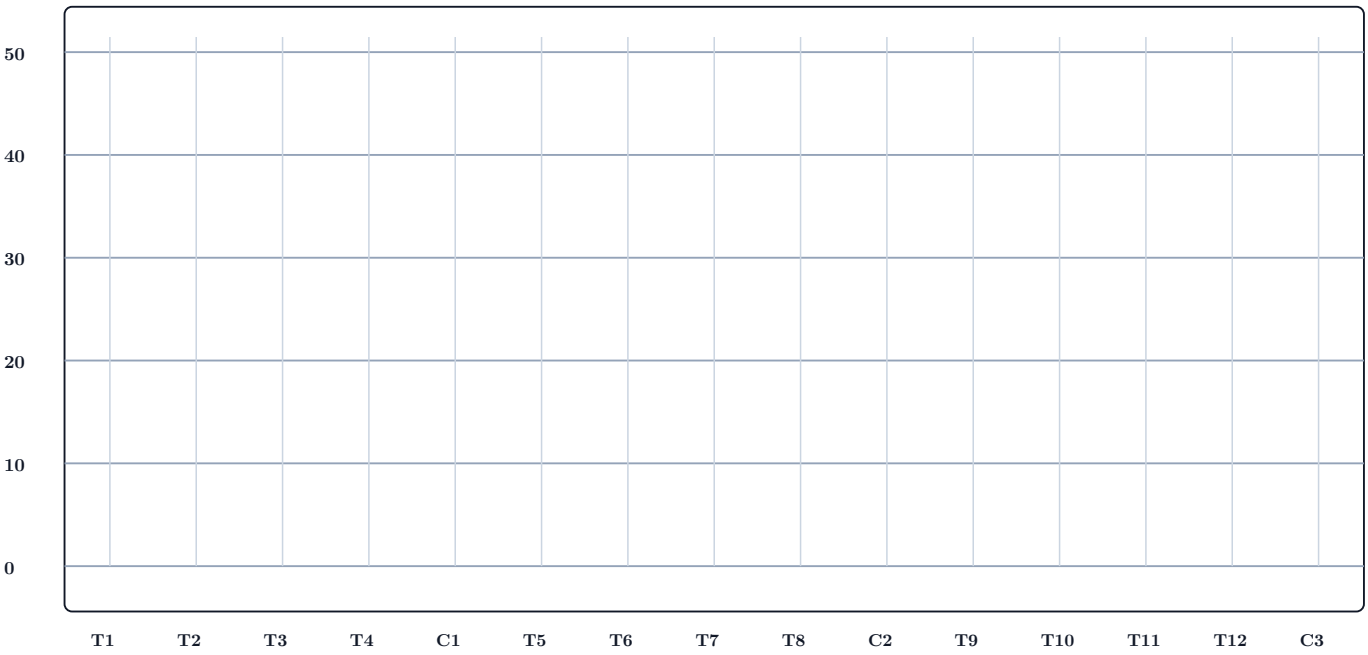
Keycap Sequence: 

Competitor Progress Graph & Certificate

Plot your score across all 15 timed trials and record your Stack-Depth Mastery certification.

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1. 15-TRIAL SCORE PROGRESSION GRAPH (PLOT YOUR SCORE FROM 0 TO 50 POINTS)



OFFICIAL CERTIFICATE OF RPN STACK PROFICIENCY

This certifies that: _____

has completed the 15 timed sprints in the StackCalc Stack Speed & Fluency Trials, demonstrating numerical accuracy, inside-out operand sequencing, and strict 4-register (T, Z, Y, X) stack depth discipline.

[ ] Level 1 Stack Badge (Single Register X) **[ ] Level 2 Stack Badge (Registers X & Y)**

[ ] Level 4 Master Badge (Full T-Z-Y-X Stack) **Highest Trial Score:** _____ / **50 pts**

Coach / Teacher Signature: _____ Date: _____